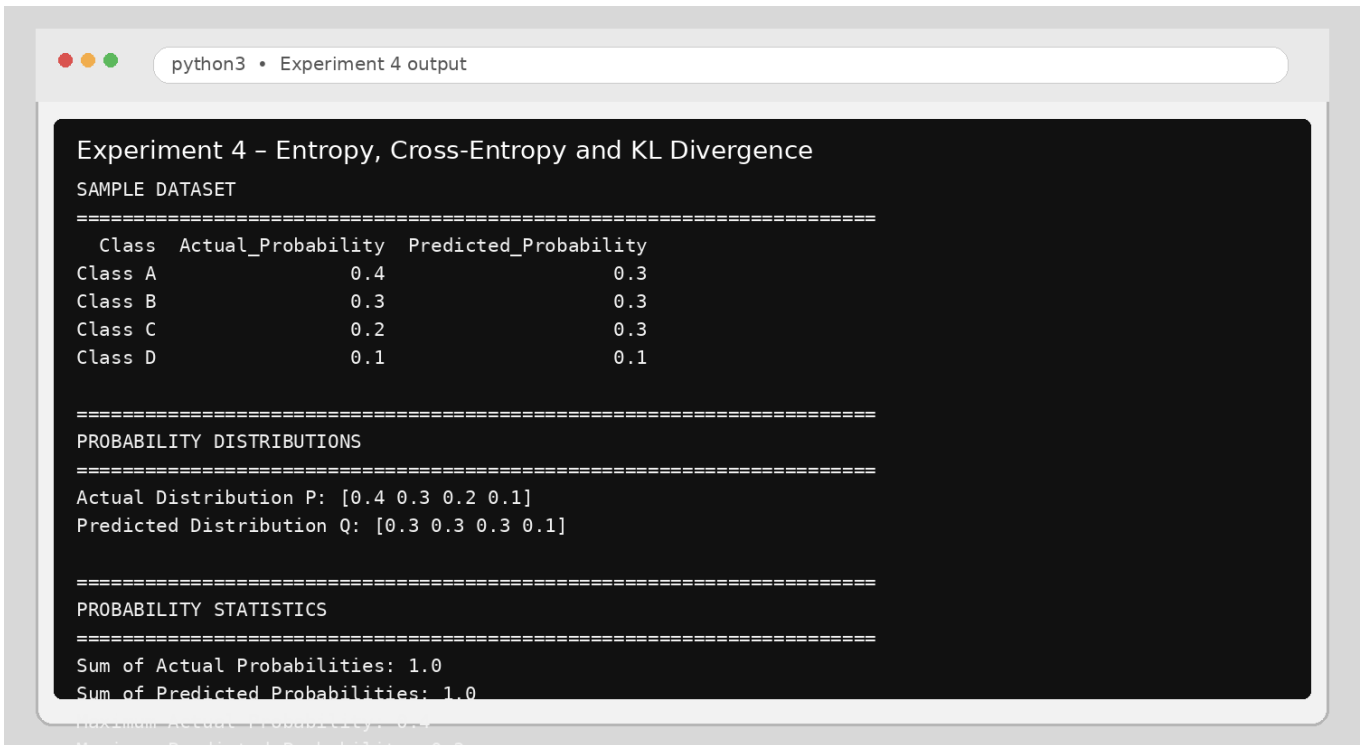


Experiment 4 – Entropy, Cross-Entropy and KL Divergence

CO1 • NumPy, SciPy, Matplotlib

A terminal window titled 'python3 • Experiment 4 output' with a dark background and light-colored text. The output displays the title of the experiment, followed by a sample dataset table comparing actual and predicted probabilities for four classes. Below the table, it shows the probability distributions P and Q as arrays. Finally, it calculates the sum of probabilities for both distributions, both resulting in 1.0.

```
python3 • Experiment 4 output

Experiment 4 – Entropy, Cross-Entropy and KL Divergence
SAMPLE DATASET
=====
  Class  Actual_Probability  Predicted_Probability
Class A           0.4           0.3
Class B           0.3           0.3
Class C           0.2           0.3
Class D           0.1           0.1

=====
PROBABILITY DISTRIBUTIONS
=====
Actual Distribution P: [0.4 0.3 0.2 0.1]
Predicted Distribution Q: [0.3 0.3 0.3 0.1]

=====
PROBABILITY STATISTICS
=====
Sum of Actual Probabilities: 1.0
Sum of Predicted Probabilities: 1.0
```

Figure 1: Sample dataset, probability distributions and probability statistics.



Experiment 4 - Entropy, Cross-Entropy and KL Divergence

ENTROPY

Entropy of Actual Distribution P: 1.8464

CROSS-ENTROPY

Cross-Entropy between P and Q: 1.8955

KL DIVERGENCE

KL Divergence between P and Q: 0.049

DETAILED EVALUATION TABLE

Class	Actual_P	Predicted_Q	Entropy_Contribution	Cross_Entropy_Contribution	KL_Contribution
Class A	0.4	0.3	0.5288	0.6948	0.166
Class B	0.3	0.3	0.5211	0.5211	0.000
Class C	0.2	0.3	0.4644	0.5272	0.117
Class D	0.1	0.1	0.3333	0.3333	0.000

Figure 2: Entropy, Cross-Entropy, KL Divergence and detailed evaluation table.



Experiment 4 – Entropy, Cross-Entropy and KL Divergence

EVALUATION SUMMARY

```
=====
Metric  Value
Entropy 1.8464
Cross-Entropy 1.8955
KL Divergence 0.0490
```

FINAL RESULT

```
=====
Entropy, Cross-Entropy and KL Divergence were successfully
calculated using NumPy and SciPy.
```

The experiment demonstrated:

1. Sample probability dataset preparation
2. Actual probability distribution
3. Predicted probability distribution
4. Entropy calculation
5. Cross-Entropy calculation
6. KL Divergence calculation
7. Individual contribution analysis
8. Statistical evaluation
9. Probability distribution comparison
10. Metric visualization

The results were successfully analyzed and visualized
using Matplotlib.

Figure 3: Evaluation summary and final experiment result.

Metric Visualization

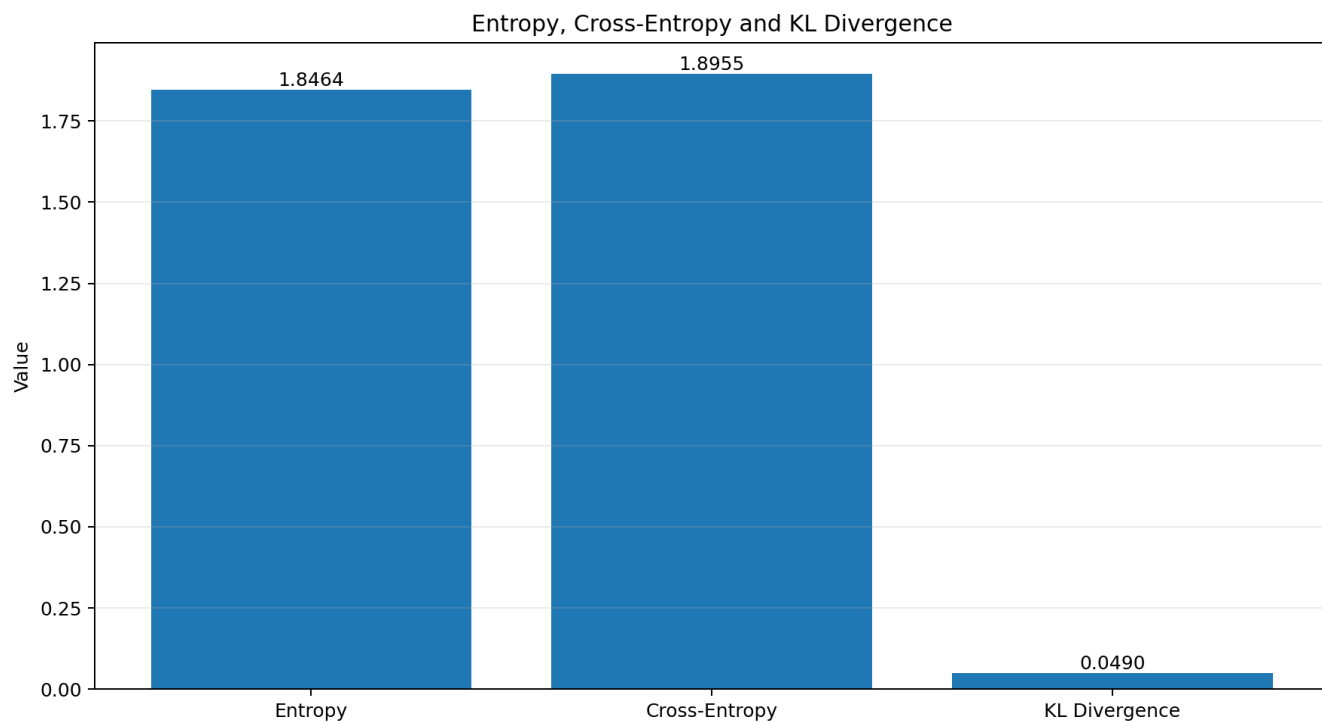


Figure 4: Comparison of the calculated entropy, cross-entropy and KL divergence values.